Decentralized Federated Anomaly Detection in Mobile Networks

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***Abstract*—Networks face growing challenges such as Distributed Denial-of-Service (DDoS) attacks, malware propagation, unauthorized access and irregular traffic patterns. Traditional centralized or static anomaly detection methods often struggle with scalability, latency and privacy concerns, making them less effective in dynamic mobile environments. This research introduces a decentralized federated learning algorithm that integrates Multi-Agent Reinforcement Learning Multi-Agent Reinforcement Learning (MARL) with tree-based classification models, using XGBoost in the experimental setup to enhance anomaly detection while preserving privacy. MARL coordinates agent behavior to adapt to dynamic network conditions, promoting decentralized collaboration and resilience without raw data sharing. Each agent trains a lightweight tree-based model locally for efficient and scalable detection. Feature selection is performed collaboratively using reinforcement signals, while trust-weighted aggregation enables decentralized global learning without a central server. The system is designed to address key constraints such as intermittent communication, which is modeled through agent dropout in the experiments, as well as computational overhead and scalability in edge network environments. Experimental results using real-world intrusion detection datasets show that the proposed method achieves up to 93.5% accuracy, 86.1% precision and 85.7% F1 score, outperforming baseline methods by as much as 23.3% in F1 score under non-IID scenarios while maintaining low bandwidth and energy usage across dynamic agent participation and varying attacker models.**

***Index Terms*—Decentralized Federated Learning, Trust-weighted Aggregation, Feature Selection, Non-IID Data, Anomaly Detection, Network Security, Peer-to-Peer Communication, Edge Computing.**

I. Introduction

The exponential growth of wireless networks has enabled real-time communication across billions of devices. However, it also introduces significant challenges for network anomaly detection, which refers to the process of identifying deviations from expected network behavior that may indicate security threats, misconfigurations, or system failures. Ensuring effective anomaly detection in modern networks requires addressing concerns related to privacy, scalability and resilience, particularly in the face of threats such as malware, evasion techniques, free-riding behavior and Distributed Denial-of-Service (DDoS) attacks.

Federated Learning (FL) reduces bandwidth usage in decentralized environments, such as mobile or edge networks, by enabling local model training while supporting real-time threat detection [1].

Decentralized Federated Learning (DFL) further removes the need for a central server, allowing direct peer-to-peer collaboration [1]. However, DFL introduces issues such as communication overhead, vulnerability to adversarial models and instability under non-IID data [2], [3].

To address the challenges posed by fully decentralized federated learning in network anomaly detection—particularly in the presence of non-IID data, adversarial behaviors, and communication constraints—this work investigates the following key research questions. Each question is directly tied to a core component of the proposed system and serves to guide the design, implementation, and evaluation of the approach.

1) **Scalability in Edge Environments:** How can a fully decentralized federated learning approach reduce communication and computational overhead while maintaining high anomaly detection performance in resource-constrained network environments?
2) **Adaptive Detection Under Data Heterogeinity:** To what extent can reinforcement-driven feature selection improve detection robustness and accuracy under non-IID data distributions?
3) **Trust-Based Aggregation for Decentralized Robustness:** How does incorporating trust-weighted peer aggregation affect the performance and stability of decentralized learning compared to naive averaging schemes?
4) **Empirical Benchmarking Against Baselines:** How does the proposed MARL-XGBoost system compare to Hierarchical Federated Learning (HDFL) and other baseline methods across real-world datasets and challenging network conditions such as agent dropout and attacker variability?

The proposed Decentralized Federation MARL-XGBoost (DFX) addresses the four research questions as follows:

1) **Scalability in Edge Environments:** DFX uses a fully decentralized communication model without central coordination, exchanging model updates directly over predefined topologies. This enables lightweight, resource-aware operation for constrained devices.
2) **Robustness Under Non-IID Data:** Each agent applies an epsilon-greedy MARL policy for adaptive feature

selection, prioritizing relevant inputs based on local learning dynamics to maintain detection accuracy.

3) **Resilience to Unreliable/Adversarial Peers:** A trust-weighted aggregation scheme adjusts each peer's influence using dynamic trust scores derived from prior performance and communication reliability.
4) **Efficiency Without Accuracy Loss:** Trust-weighted peer-to-peer aggregation and reinforcement-driven feature selection minimize redundant exchanges and payload sizes, reducing communication overhead and energy use while preserving high detection accuracy.

For empirical benchmarking, the system is evaluated on three benchmark datasets under varying network conditions, including agent dropout, packet loss and attacker heterogeneity, to demonstrate detection accuracy, communication efficiency and robustness.

Its fully decentralized structure integrates adaptive feature selection, trust-weighted peer aggregation and efficient model exchange strategies to maintain scalability and robustness in edge environments. These design elements also enable DFX to adapt effectively under non-IID data distributions and benchmark competitively against established baselines such as HDFL and Logistic Regression across diverse datasets and network conditions. We propose a decentralized algorithm combining Multi-Agent Reinforcement Learning MARL and Extreme Gradient Boosting (XGBoost). MARL enables adaptive feature selection and peer interaction, while XGBoost offers efficient, accurate local detection. Experiments simulate real-world conditions including mobility, node churn, non-IID distributions and diverse attacks. Key contributions include the development of a novel decentralized algorithm for anomaly detection, the use of trust-based aggregation to enhance robustness in adversarial settings and a parameter study evaluating performance across network topologies, agent dynamics and attack scenarios.

The remainder of this paper is structured as follows. Section II discusses related work on federated and decentralized anomaly detection. Section IV introduces the proposed MARL-XGBoost framework. Section V-A describes the experimental design and Section VI concludes this paper.

II. Related Works

Effective anomaly detection is critical for maintaining network security, specifically to identify deviations indicative of cyber threats such as Distributed Denial-of-Service (DDoS) attacks and malware propagation [9]. The datasets used in this study include diverse categories of anomalies: Denial-of-Service (DoS/DDoS) attacks (e.g., SYN floods, HTTP floods) that overwhelm network resources, reconnaissance and probing attacks (e.g. port scans) that map network vulnerabilities, botnet and backdoor traffic (e.g. command-and-control traffic) that can compromise large numbers of hosts, infiltration and privilege escalation attacks, and data exfiltration and exploitation-based attacks (e.g. credential theft, fuzzers) that steal or damage information. These anomalies are drawn from the UNSW-NB15 [10], CICIDS2017 [11], and TON_IoT [12] datasets and represent both high-intensity and evasive threats relevant to real-world networks. The UNSW-NB15 dataset includes a range of modern attack types such as DoS/DDoS, reconnaissance and backdoors, capturing contemporary threat behaviors on emulated enterprise networks [10]. CICIDS2017 is widely used for intrusion detection research and features realistic network traffic mixed with diverse attack scenarios, including brute-force, botnet and infiltration attempts [11]. TON_IoT focuses on Internet-of-Things (IoT) environments, providing a mix of telemetry data and cyberattacks such as ransomware and data exfiltration that highlight vulnerabilities unique to distributed edge devices [12].

TABLE I
Comparison of Related Works.
C= Centralized, D=Decentralized.
Coordination refers to whether the system uses a central controller or peer-to-peer exchange.

Work	Method	Arch.	FL Strategy	Coord.	Features
[4]	Hierarchically Decentralized Federated Learning (HDFL)	C	D	C	Categorical
[5]	Secure Peer-to-Peer Federated Learning	D	D	D	Categorical
[6]	Self-Organizing Networks (SON)	C	D	D	Categorical
[2]	Peer-to-Peer FL for Edge Devices	D	D	D	Categorical
[7]	Survey on Decentralized FL	D	D	D	Mixed
[3]	FedADSN for Social Network Anomalies	D	D	D	Graph Anomaly
[8]	FL-based Network Anomaly Detection	D	D	D	Categorical

Traditional centralized network anomaly detection methods face limitations such as high communication costs, increased latency [13] and privacy risks.

In contrast, FL allows nodes to collaboratively train anomaly detection model by exchanging model updates rather than raw data, thereby enhancing scalability and privacy [7]. DFL has emerged as an effective distributed machine learning approach, enabling collaborative training and model aggregation across decentralized nodes without a central server [7]. Compared to Centralized Federated Learning, which depends on a certain orchestrator, DFL distributes responsibilities among nodes, improving scalability, resilience to single points of failure and data privacy through localized data processing [1].

Despite these advantages, DFL introduces unique challenges such as increased communication overhead, the need for adaptive resource allocation [8] and complexities in managing dynamic network topologies, especially in resource-constrained environments such as wireless networks. Recent studies have proposed adaptive DFL solutions, including Peer-to-Peer (P2P) algorithms to enhance fault tolerance and scalability through direct node-to-node communication [2]. Hierarchical decentralized approaches optimize inter and intra-cluster communications, effectively reducing latency and energy consumption [4].

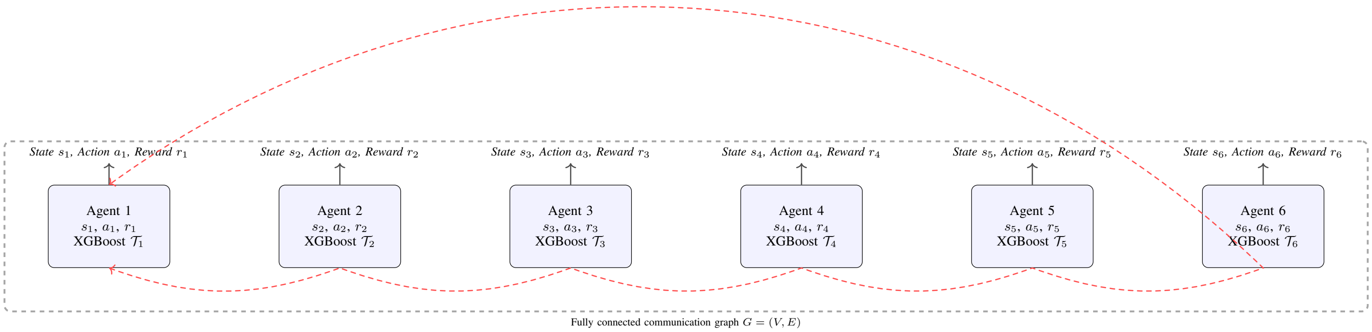


Fig. 1. Decentralized MARL-XGBoost architecture with peer-to-peer connectivity.

One such example is the Hierarchical Decentralized Federated Learning (HDFL) algorithm, which employs local D2D (device-to-device) consensus within clusters while relying on a central base station for inter-cluster aggregation [4]. While often referred to as "decentralized", HDFL is more accurately described as a hybrid architecture, since its global communication still relies on a central server [4]. This distinction is crucial for evaluating decentralized resilience and scalability and justifies its use in our study as a centralized baseline.

However, existing methods often rely on static optimization strategies that target fixed communication schedules and global model accuracy, limiting their effectiveness in unpredictable Real-World (RW) network conditions, such as intermittent connectivity and heterogeneous resource availability [3]. To overcome these limitations, recent works have explored MARL to dynamically adapt DFL to real-time network conditions. MARL facilitates cooperative optimization of model aggregation frequencies and participant selection, directly responding to network changes [6].

Compared to traditional reinforcement learning (RL), MARL demonstrates superior performance in environments demanding collaborative decision-making among distributed agents [4]. Furthermore, integrating Extreme Gradient Boosting XGBoost, a computationally efficient and lightweight algorithm [14], provides enhanced network anomaly detection capabilities suitable for constrained environments [15]. The effectiveness of reinforcement learning for adaptive optimization in DFL algorithms is well-established. Recent studies have validated MARL's ability to dynamically improve detection accuracy [16] and resource efficiency under varying network conditions [16]. Similarly, the computational efficiency and effectiveness of XGBoost have been demonstrated across real-world scenarios [3], significantly outperforming conventional methods in network anomaly detection tasks within resource-limited environments [17]. In contrast to previous approaches that largely rely on static optimization methods, this research integrates MARL and XGBoost, introducing adaptive real-time model exchange policies coupled with efficient network anomaly detection. By addressing the limitations related to dynamic network conditions, this combined approach provides a scalable, privacy-preserving and resource-aware solution specifically tailored for modern network environments, effectively bridging existing gaps within current literature. Motivated by the limitations identified in the reviewed approaches, we propose a decentralized MARL-XGBoost algorithm. The technical design and implementation are described in Sections III and V-A.

III. Decentralized Federated MARL-XGBoost

A. System Model

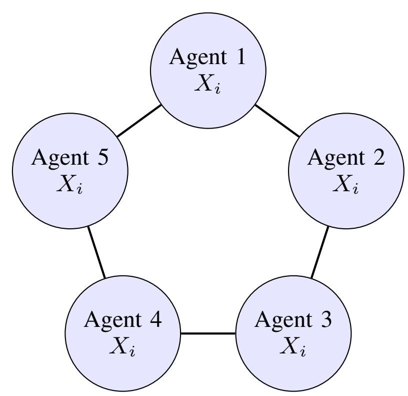


Fig. 2. Ring topology for decentralized MARL-XGBoost Collaboration

We consider a decentralized anomaly detection system deployed in a simulated network setting, where a set of agents
$\mathcal{A} = \{g_1, g_2, \ldots, g_N\}$
operate on distributed edge devices. Communication is modeled as an undirected graph $G = \{V, E\}$, where each edge denotes a peer-to-peer link with associated bandwidth, latency and packet loss to reflect mobile communication dynamics.

Figure 1 illustrates the overall Decentralized Federated MARL-XGBoost (DFX) framework. In this fully decentralized design, each agent observes its local environment, trains a local XGBoost model enhanced by Multi-Agent Reinforcement

Learning (MARL), and exchanges model updates with peers across the communication graph $G = (V, E)$.

These peer-to-peer interactions enable model improvements without reliance on a central aggregator, reducing single points of failure and preserving data privacy. Trust-weighted aggregation mechanisms ensure that unreliable or malicious agents have limited influence, while adaptive feature selection and neighbor collaboration allow agents to cope with non-IID data and dynamic network conditions. In this context, trust-weighted aggregation refers to the process of assigning greater influence to updates from trustworthy neighbors, those with consistent performance and reliable data, while reducing the impact of less reliable peers. Trust scores are updated dynamically based on metrics such as local F1-score, communication reliability and historical consistency. This ensures that model updates from unreliable or malicious agents do not degrade the learning process and makes the system robust against adversarial behaviors. This framework forms the foundation of the proposed anomaly detection approach evaluated in later sections. Among the evaluated topologies, the ring topology is particularly advantageous for fully decentralized networks.

Each agent communicates only with two immediate neighbors, which reduces communication overhead and makes bandwidth requirements predictable as the network scales. This structure also improves fault tolerance because failures in one part of the network do not isolate other agents, unlike star-based topologies. Compared to fully connected networks, which incur quadratic communication costs as the number of agents increases, the ring topology provides a scalable and energy-efficient alternative while maintaining sufficient connectivity for information propagation.

Figures 1 and 2 illustrate the proposed Decentralized Federated MARL-XGBoost (DFX) framework and its representative communication topologies. Figure 1 shows the end-to-end DFX architecture, where each agent learns locally using MARL-enhanced XGBoost and exchanges model updates with peers across a fully connected communication graph $G = (V, E)$. Figure 2 highlights the *ring topology*, one of the core topologies supported by DFX. The ring topology offers a key advantage in our proposed approach: it ensures balanced communication overhead across all agents, as each agent is only required to communicate with its two direct neighbors. This design reduces bandwidth usage compared to a fully connected graph while maintaining robust peer-to-peer connectivity and resilience against agent failures. Additionally, the structured connectivity allows more deterministic propagation of model updates, improving stability and convergence in dynamic mobile network environments. Each agent g_i owns a local dataset $D_i = (X_i, y_i)$, where i indexes the agents and n_i denotes the number of data points (samples) in agent i's dataset. $X_i \in \mathbb{R}^{n_i \times f}$ contains flow or packet-level features (e.g., protocol, duration, byte count) and $y_i \in \{0, 1\}^{n_i}$ indicates benign or anomalous traffic.

Agents train local tree-based classifiers and collaborate in a fully decentralized manner by exchanging trust-weighted parameter updates, referred to as deltas ($\Delta\theta$), with their neighbors in G.

The deltas represent the change in model parameters since the previous round and are applied by neighboring agents to incrementally update their own models. This approach avoids the redundancy of transmitting full model weights while preserving communication efficiency.

Raw data is never shared, ensuring privacy. Agent participation is dynamic, reflecting real-world edge scenarios where devices may enter or leave the network over time, this behavior is explicitly evaluated in the experimental section.

Each agent observes its local performance and communication metrics, such as F1-score, latency and packet loss, which form the state inputs to a MARL policy. This policy selects feature subsets (denoted as $\mathcal{F}_i \subseteq \mathcal{F}$ for agent g_i) and determines which neighbors to collaborate with using ϵ-greedy strategy with decay.

Network conditions influence the reward signal, which adjusts trust values used in aggregating updates from neighbors. This process enables agents to learn adaptively from reliable peers and maintain robustness under non-IID, lossy and dynamic network conditions.

B. Problem Formulation

We aim to detect network anomalies that occur simultaneously across multiple agents(i.e. *distributed anomalies*) with high classification performance and minimal communication overhead in a fully decentralized, multi-agent setting.

The datasets used (CICIDS2017, UNSW-NB15 and TON_IoT) contain coordinated attack scenarios such as botnets and distributed denial-of-service (DDoS), which justify this definition. Each agent $g_i \in \mathcal{A}$ trains a local classifier h_{θ_i},parameterized by weights θ, using only its local dataset.

To mitigate the impact of class imbalance, each agent applies a weighting coefficient μ_i defined as

$$\mu_i = \frac{\text{number of benign samples}}{\text{number of anomalous samples}}$$

to balance the loss function during training.

To address class imbalance, we introduce a weighting coefficient μ_i, defined as

$$\mu_i = \frac{n_i}{2 \cdot \|y_i\|_1}$$

The local loss function minimized by agent a_i is then defined as

$$\mathcal{L}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \ell(h_{\theta_i}(x_{ij}), y_{ij}; \mu_i)$$

where $\ell(\cdot)$ is a scale-sensitive binary cross-entropy loss function, defined as

$$\ell(\hat{y}, y; \mu_i) = -\mu_i \cdot y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

and $\hat{y} = h_{\theta_i}(x)$ denotes the predicted probability that the instance x is anomalous.

The $F1_i$ score for agent g_i is computed as:

$$F1_i = \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

where precision and recall are derived from the agent's local predictions on its dataset.

The global performance objective seeks to maximize average detection quality across all agents while penalizing the communication cost incurred during collaborative updates.

The MARL-XGBoost decentralized learning process is summarized in Algorithm 1, where each agent iteratively trains a local XGBoost model, selects a feature subset via an epsilon-greedy policy and exchanges model updates with neighbors in a fully peer-to-peer manner. Trust-weighted aggregation is used to combine received updates, where each neighbor's contribution is scaled based on a dynamically updated trust score that reflects past performance and communication reliability. The agent's model is updated using a weighted sum of parameter deltas and the trust score is refined each round based on a reward signal incorporating F1-score and communication cost. The nested for loops represent the repeated process of local model training, peer-to-peer aggregation and policy adjustment over communication rounds and across neighboring agents.

Algorithm 2 outlines the HDFL baseline process, which follows a structured, hierarchical federated learning approach. Each agent trains a local model and communicates within a predefined cluster, where updates are aggregated using simple averaging without any trust mechanism. Feature subsets are selected using the same epsilon-greedy strategy for consistency, but all peer contributions are treated equally, regardless of reliability or performance history. The for loop at line 5 iterates over all participating agents, enabling each to perform local model training within its data partition. The for loop at line 7 iterates over the communication rounds, during which peer-to-peer exchanges occur and models are updated based on uniform neighbor averaging. This structure ensures that every agent trains locally in each round and then synchronizes its parameters with neighbors, following a non-adaptive coordination strategy without trust weighting. This baseline reflects a non-adaptive coordination strategy that lacks the robustness of trust-aware aggregation and is used for the comparative benchmarking against the proposed decentralized system.

Algorithm 1: Decentralized Federated Anomaly Detection with Trust-Weighted Aggregation

// **Input:** Local data (X_i, y_i) for each agent i, communication graph $G = (V, E)$, initial trust scores $t_i^{(0)}$, learning rate η, decay factor β, exploration rate ϵ

// **Output:** Updated local classifiers for each agent

foreach *agent* $i \in V$ **do**
    // Train initial XGBoost model on local data

foreach *round* $t = 1, \ldots, T$ **do**
    foreach *agent* $i \in V$ **do**
        // Select feature subset $f_i^{(t)}$ via ϵ-greedy strategy
        // Exchange model parameters $\theta_j^{(t)}$ with neighbors $j \in N(i)$
        // Compute trust weights

$$w_{ij}^{(t)} = \frac{t_j^{(t)}}{\sum_{k \in N(i)} t_k^{(t)}}$$

        // Update model using trust-weighted aggregation

$$\theta_i^{(t+1)} \leftarrow \theta_i^{(t)} + \eta \sum_{j \in N(i)} w_{ij}^{(t)} (\theta_j^{(t)} - \theta_i^{(t)})$$

        // Evaluate updated model and compute reward $r_i^{(t)}$
        // Update trust value

$$t_i^{(t+1)} = \beta t_i^{(t)} + (1 - \beta) r_i^{(t)}$$

IV. System Overview

Our proposed method combines Multi-Agent Reinforcement Learning (MARL) with XGBoost to achieve adaptive, decentralized anomaly detection in dynamic and heterogeneous network environments. This design explicitly addresses challenges of non-IID data distributions, intermittent connectivity and the need for high detection performance while minimizing communication overhead. The system operates over a time-varying communication graph $G = \{V, E\}$, where each node $v_i \in V$represents an agent and each edge $(v_i, v_j) \in E$ denotes a peer-to-peer link. Links have associated latency, bandwidth and packet loss values, reflecting realistic network dynamics such as mobile edge devices joining or leaving the system. This time variability is evaluated in the experimental setup by simulating agent churn and fluctuating link quality. Each agent g_i maintains a local state vector $s_i^{(t)}$ at round t that captures the information needed for decision-making. The vector contains three elements. The first element is a binary mask that shows which subset of features is currently selected for model training.

The second element records recent detection performance metrics such as the local F1-score and classification loss. The third element includes communication statistics, for example observed latency, bandwidth usage and packet loss from recent interactions with neighbors. These elements together allow the agent to evaluate both its predictive performance and the quality of its communication links. The state vector is updated at every round so that agents can adapt their decisions to changing network conditions and data distributions.

At each round, the agent g_i chooses an action $a_i^{(t)}$ based on its current state vector $s_i^{(t)}$. The action defines two decisions.

The first decision is which subset of features will be used for model training during the current round. The second

Algorithm 2: Decentralized Federated Anomaly Detection with Baseline Averaging

// **Input:** Local data (X_i, y_i) for each agent i, communication graph $G = (V, E)$, learning rate η, exploration rate ϵ

// **Output:** Updated local classifiers for each agent

foreach *agent* $i \in V$ **do**
  // Train initial XGBoost model on local data

foreach *round* $t = 1, \ldots, T$ **do**
  foreach *agent* $i \in V$ **do**
    // Select feature subset $f_i^{(t)}$ via ϵ-greedy strategy
    // Exchange model parameters $\theta_j^{(t)}$ with neighbors $j \in N(i)$
    // Update model using simple averaging

$$\theta_i^{(t+1)} \leftarrow \theta_i^{(t)} + \eta \frac{1}{|N(i)|} \sum_{j \in N(i)} (\theta_j^{(t)} - \theta_i^{(t)})$$

    // Evaluate updated model and compute reward $r_i^{(t)}$

decision is which neighbors in the communication graph the agent will exchange updates with. The action is selected using ϵ-greedy strategy with decay, which means the agent balances exploration of new actions and exploitation of actions that have produced good results in the past. Over time, the probability of exploring new actions decreases as the agent learns which actions yield higher rewards. The reward signal $r_i^{(t)}$ is designed to guide each agent g_i toward improving both detection performance and communication efficiency. At the end of each round t, the agent evaluates the outcome of its action $a_i^{(t)}$ using a combined metric.

The reward is computed as a weighted sum of the agent's F1-score and a penalty terms for communication cost, which accounts for factors such as bandwidth consumption and latency experienced during peer-to-peer exchanges. This formulation encourages agents to achieve high detection accuracy while minimizing unnecessary communication. Over time the reward signal helps agents refine their policies by favoring actions that yield a balanced trade-off between quality and resource efficiency.

Each agent g_i maintains a local XGBoost model h_{θ_i} because of its capability to handle high-dimensional and imbalanced datasets effectively. XGBoost builds an ensemble of decision trees using gradient boosting, which is particularly suitable for structured flow-level data used in this work. Training is performed exclusively on the agent's local dataset and raw data is never transmitted, preserving privacy across the system. After training, each agent exchanges parameter updates, represented as deltas $\Delta\theta$, with its selected neighbors.

These deltas capture the change in model parameters since the previous round and allow for efficient communication without sending full model weights.

The updates are combined using a trust-weighted aggregation scheme in which each neighbor's contribution is weighted according to a dynamically updated trust score. This score reflects how much the neighbor's past updates improved the agent's detection performance. By prioritizing reliable peers, the system reduces the impact of untrustworthy updates and maintains robustness in non-IID, lossy and dynamic network conditions. The entire learning process proceeds as a decentralized loop, allowing agents to iteratively refine both their models and decision-making policies. At each round t, every agent g_i begins by observing its current state vector $s_i^{(t)}$, selecting an action $a_i^{(t)}$, and updating its local XGBoost model based on the selected feature subset and training data.

Agents then exchange parameter deltas $\Delta\theta$ with the selected neighbors, applying trust-weighted aggregation to incorporate updates from reliable peers.

Following this update, the reward $r_i^{(t)}$ is computed to evaluate the round's performance, directly influencing the trust scores and guiding future decisions. This iterative cycle enables agents to adapt dynamically to changing network conditions and adversarial behaviors while maintaining complete decentralization and strong privacy guarantees.

The decentralized learning loop continues for a fixed number of rounds T or until convergence is reached. In each iteration, agents refine their local models based on newly observed states, selected actions and received updates from neighbors.

This iterative process enables agents to improve anomaly detection performance while maintaining robustness to non-IID data distributions and intermittent network conditions. Because updates are exchanged only with selected peers and no raw data is shared, the system preserves data privacy and reduces communication overhead compared to centralized methods. The following section details the experimental setup used to evaluate the proposed framework and its performance under various network conditions and attack scenarios.

V. Performance Evaluation

A. Decentralized Federated Benchmark Performance Test

We refer to our proposed method as **DFX** (Decentralized Federated MARL-XGBoost). DFX represents a fully decentralized anomaly detection framework that combines Multi-Agent Reinforcement Learning (MARL) with XGBoost classifiers for robust local model updates, adaptive feature selection, and trust-weighted peer-to-peer aggregation. We compare the performance of our fully decentralized MARL-XGBoost system with the following baseline:

- **Hierarchical Federated Learning (HDFL):** Agents are clustered into cells with intra-cell and inter-cell aggregation phases. HDFL represents a structured federated approach to improve scalability and reduce communication overhead.

- **Logistic Regression:** Each agent independently trains a logistic regression classifier on its local data and applies naive averaging of model outputs across neighbors without reinforcement learning, feature selection, or trust-weighted aggregation. This baseline simulates an uncoordinated decentralized system with minimal communication and no adaptive behaviour.

B. Testbed and Simulation Environment

We evaluate MARL-XGBoost, decentralized anomaly detection system combining Multi-Agent Reinforcement Learning and XGBoost classifiers. Experiments use UNSW-NB15, CICIDS2017 and TON_IoT datasets, covering diverse network traffic and attacks. Preprocessing involves label binarization, standard scaling and correlation-based feature filtering. Dirichlet-based partitioning simulates non-IID data, with stratified fallback when class imbalance occurs. The experiments were conducted on a MacBook Pro equipped with an Apple M1 chip, 16 GB of RAM and macOS Ventura 13.6. The software environment consisted of Python 3.11, the VSCode IDE and widely used libraries such as NumPy 1.26.4, Pandas 2.2.2, Scikit-learn 1.5.0, XGBoost 2.1.0, Matplotlib 3.9.2, Seaborn 0.13.2 and imbalanced-learn 0.12.3. Agents operate ring, star, fully connected and random topologies, with dynamic join/leave behavior emulating real-world network conditions.

C. Simulation Parameters and Performance metrics

Each agent computes its F1-score by combining its precision and recall using the standard harmonic mean formula. Where precision and recall are derived from local predictions on each agents' dataset. Let C denote the composite network cost, which includes factors such as latency, bandwidth usage and packet loss across the system. The Optimization goal is then

$$\min_{\theta_1,\ldots,\theta_N} \left(\frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_i + \lambda \cdot C \right)$$

where $\mathcal{L}_i$ is the local loss for agent a_i, C denotes the average communication cost (latency, bandwidth, and packet loss), and $\lambda \geq 0$ is a regularization parameter controlling the trade-off between training quality and communication efficiency.

D. Aggregate Global Model Formulation

In the proposed decentralized anomaly detection algorithm, each agent trains a local XGBoost classifier using a subset of features from its local dataset to reduce computation and communication overhead. To capture real-world heterogeneity, agents operate on unique non-IID data partitions generated through Dirichlet sampling in each simulation. Local XGBoost models are trained at each agent using a histogram-based algorithm suitable for structured data and constrained resources.

Feature selection is dynamically performed using an ϵ-greedy strategy (see Table II for the initial ϵ value and decay rate). After training, agents exchange models with neighbors defined by the network topology and perform trust-weighted aggregation to prioritize reliable sources.

TABLE II
SIMULATION SETTINGS AND EVALUATION METRICS

Parameter / Category	**Value / Description**
Agent Count	5–10 (with dynamic joining/leaving per round)
Datasets Used	UNSW-NB15, CIC-IDS2017, TON_IoT
Communication Rounds	3 rounds per topology per mode (trust/baseline)
Network Topologies	Ring, Star, Fully Connected, Random
Classifier	XGBoost (used in both trust-weighted and naive baseline modes)
Aggregation Strategy	Trust-weighted aggregation (main) or naive averaging (baseline)
Imbalance Handling	Adaptive class weighting + SMOTE (with fallbacks)
Data Distribution	Non-IID (Dirichlet $\alpha \in \{0.1, 0.3, 0.5, 1.0, 2.0\}$)
Feature Selection	Agent-local epsilon-greedy with initial $\epsilon = 0.3$, decay rate = 0.95 per round
Oversampling Strategy	Agent-specific SMOTE with class checks
Attack Scenarios	DoS, Reconnaissance, Exploits, Fuzzers
Energy Awareness	Simulated for training and communication
Detection Metrics	Accuracy, Precision, Recall, F1-score
Network Metrics	Bandwidth (Mbps), Latency (ms), Packet Loss (%)

Energy usage is logged per round to assess communication and computation efficiency.

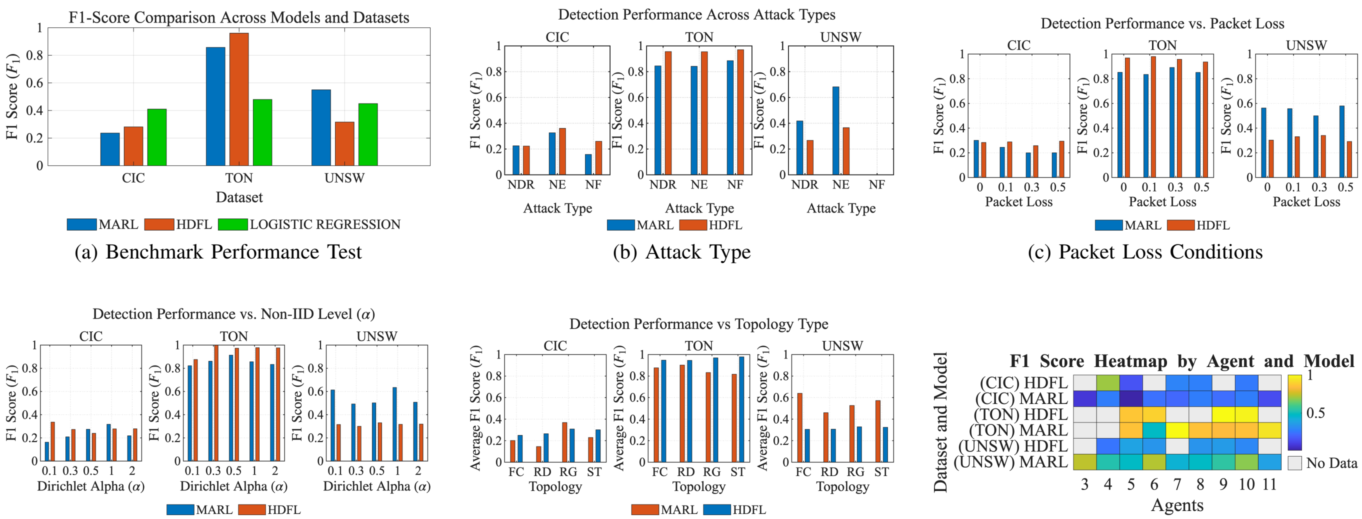
Experiments span four topologies:

Ring, Star, Fully Connected and Random, to evaluate performance under varying structural constraints. The Random topology represents a random mesh network where each agent is connected to a randomly selected subset of peers, simulating unstructured and dynamic network conditions such as those found in mobile ad hoc or IoT networks. We use three benchmark intrusion datasets, CICIDS2017, UNSW-NB15 and TON_IoT. Data preprocessing includes null removal, correlation filtering, standardization and label binarization, followed by stratified train-test splitting.

To benchmark our decentralized MARL-XGBoost system, we compare against two alternatives,as introduced in Section V-A, a centralized Hierarchical Federated Learning (HDFL) baseline using logistic regression with intra/inter-cluster aggregation and a logistic regression setup with simple averaging and no reinforcement mechanism. Detection metrics include accuracy, precision, recall and F1-score and network metrics include bandwidth, latency and packet loss

E. Results Analysis

This section presents an analysis of the experimental results obtained from evaluating the proposed MARL-XGBoost algorithm against a Hierarchically Decentralized Federated Learning baseline across varying network conditions. The performance of each method was assessed using F1-score, averaged across datasets and experimental runs, to ensure robustness and generalizability. The results are grouped into four key areas, each answering a core research question. The datasets on devices are non-IID and imbalanced, generated by partitioning using Dirichlet allocation [2].



(a) Benchmark Performance Test (b) Attack Type (c) Packet Loss Conditions

(d) Non-IID level (α) (e) Network Topology (f) Agent-level view

Fig. 3. F1-score comparison across six experiment conditions.

1) Comparative Detection Performance Across Data and Attack Conditions: Figure 3a shows that MARL-XGBoost delivers more stable F1 performance across datasets and attack types than both HDFL and Logistic Regression baselines. Although HDFL occasionally outperforms MARL-XGBoost, as in exploit detection on the CICIDS2017 dataset, MARL-XGBoost achieves the highest overall F1 scores on the complex UNSW-NB15 dataset and exhibits lower variability across attack categories. This stability stems from its MARL-driven feature selection and trust-weighted aggregation, where agents assign greater influence to peer updates according to dynamic trust scores reflecting prior detection accuracy and communication reliability. These mechanisms enable effective adaptation under heterogeneous nous data distributions and unreliable peers, producing consistent results even in dynamic network conditions. In contrast, Logistic Regression lacks adaptive mechanisms and performs worse on most datasets, reinforcing the value of collaborative learning. Overall, the results highlight MARL-XGBoost's practical advantages for fully decentralized anomaly detection in edge network enviroments.

2) Detection Performance by Attack Type: Figure 3b highlights how MARL-XGBoost and HDFL perform across different attack categories. MARL-XGBoost maintains higher or comparable F1-scores for most attack types, showing particularly strong resilience in detecting complex attacks such as reconnaissance and multi-stage exploits. The results emphasize the framework's ability to generalize across attack patterns, which is critical in dynamic and evolving threat landscapes.

3) System Robustness Under Packet Loss: As shown in Figure 3c, MARL-XGBoost maintains higher detection accuracy under increasing packet loss compared to HDFL. Its adaptive communication strategy mitigates unreliable connections, outperforming HDFL beyond a 0.1 loss rate. This robustness comes from its trust-weighted aggregation and decentralized peer-to-peer updates, which allow agents to down-weight unreliable peers and recover information from multiple neighbors when packet loss occurs.

4) Adaptation to Non-IID Data Distributions: Figure 3d illustrates how MARL-XGBoost and HDFL handle non-IID data distributions. MARL-XGBoost performs comparably to HDFL when data heterogeneity increases, thanks to its MARL-driven feature selection and trust-weighted aggregation mechanisms. HDFL performs better under balance distributions (higher α), while MARL-XGBoost outperforms or matches HDFL as α decreases, simulating more realistic, highly non-IID scenarios. At $\alpha = 0.3$, HDFL scores 0.552 compared to MARL-XGBoost's 0.524. This suggests that while both systems degrade under non-IID conditions, MARL-XGBoost adapts better due to its reinforcement learning-based communication and aggregation strategies, addressing anomaly detection performance under non-IID data.

5) Impact of Network Topology: Figure 3e compares detection performance across different topologies (Ring, Star, Fully Connected, Random Mesh). The random topology represents a mesh network where each agent is connected to a randomly selected subset of peers, simulating unstructured and dynamic network conditions such as those founds in mobile and ad hoc or IoT networks. While the fully connected topology achieves slightly higher F1-scores, the performance gain over the ring topology is marginal relative to the significant increase in bandwidth and energy cost. Although the fully connected topology offers the highest theoretical connectivity, its feasibility in fully decentralized environments is limited by quadratic communication overhead as the number of agents increases. The ring topology achieves comparable detection accuracy while requiring each agent to communicate with only two neighbors, making it most practical for large-scale deployment.

6) Data Availability Across Configurations: Figure 3f shows average F1-score of MARL-XGBoost and HDFL across datasets and agent counts. ”No Data” cells indicate configurations where partitioning failed to ensure class diversity or sample size, preventing reliable training. MARL-XGBoost maintains more scalable performance across agent counts, demonstrating scalability and robustness even with sparse data.

VI. Conclusion

This work presented a decentralized MARL-XGBoost framework for network anomaly detection, combining MARL's adaptive decision-making with XGboost's efficiency to enhance accuracy, scalability and resource awareness. Evaluations on UNSW-NB15, CICIDS2017 and TON_IoT showed superior performance to baselines across diverse conditions. Future work will refine communication efficiency, extend to additional datasets and explore real-time deployment.

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